

KAUNAS, Lithuania

WMO: 266290

Lat: 54.8834N

Lon: 23.8349E

Elev: 253

StdP: 14.56

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.7	4.2	-6.2	4.0	-1.3	-0.6	5.4	4.8	24.2	30.9	22.4	29.0	4.2	160	0.570

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.9	83.5	67.8	80.3	66.0	77.2	64.5	70.0	79.5	68.0	76.6	66.2	74.4	8.0	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.8	99.5	74.3	64.9	92.9	72.0	63.1	87.1	70.3	34.1	79.1	32.4	76.5	31.0	74.4	78.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.4	18.1	16.1	-7.3	88.6	7.1	3.3	-12.4	90.9	-16.5	92.9	-20.5	94.7	-25.7	97.1		
			-7.7	73.1	7.0	2.2	-12.7	74.7	-16.8	76.0	-20.7	77.2	-25.7	78.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.5	25.5	27.0	34.0	45.9	55.0	60.7	65.2	64.1	55.6	45.3	36.9	29.3
	DBStd	16.01	10.59	10.26	7.81	7.87	7.24	5.79	5.58	5.28	5.87	6.70	7.40	9.59
	HDD50	3332	760	644	497	171	31	1	0	0	17	176	394	641
	HDD65	7296	1225	1064	961	573	321	155	67	82	287	611	844	1106
	CDD50	1678	0	0	1	48	184	321	472	437	185	30	0	0
	CDD65	168	0	0	0	1	9	25	75	54	4	0	0	0
	CDH74	1436	0	0	0	11	98	220	617	456	34	0	0	0
	CDH80	301	0	0	0	0	10	32	152	104	3	0	0	0
Wind	WSAvg	7.9	9.4	9.1	8.6	7.8	7.2	6.9	6.3	6.3	7.0	8.2	9.1	9.4
Precipitation	PrecAvg	24.8	1.6	1.2	1.4	1.5	2.1	2.7	3.2	3.0	2.2	2.2	2.0	1.9
	PrecMax	33.3	3.6	2.7	3.2	3.4	6.0	5.4	7.9	6.9	4.4	6.6	4.8	3.0
	PrecMin	17.6	0.1	0.2	0.4	0.2	0.4	0.3	0.2	0.2	0.3	0.2	0.4	0.6
	PrecStd	3.6	0.8	0.6	0.7	0.8	1.1	1.2	1.8	1.7	1.0	1.4	0.9	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.0	45.8	58.0	75.4	81.3	84.2	88.9	88.0	78.4	65.2	53.4	48.0
		MCWB	43.6	42.8	47.4	58.8	65.1	67.9	69.5	69.2	63.8	57.3	50.5	46.0
	2%	DB	42.0	43.1	52.2	69.5	76.7	79.9	84.3	82.7	72.6	60.6	50.4	44.8
		MCWB	40.2	40.6	44.5	55.3	61.6	65.7	68.7	66.6	61.5	55.4	48.3	43.1
	5%	DB	39.5	40.5	47.8	64.3	73.0	76.4	80.7	79.1	68.6	57.4	48.0	42.4
		MCWB	38.0	38.6	42.6	52.2	59.8	63.9	67.2	64.9	59.7	53.5	46.3	41.0
	10%	DB	37.1	38.5	44.0	59.8	69.3	72.9	77.2	75.5	65.4	54.8	46.3	40.0
		MCWB	36.1	36.9	40.3	49.9	58.1	62.0	65.8	63.6	58.3	51.8	44.8	38.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.9	43.5	48.8	60.1	67.6	70.7	73.7	73.4	65.5	59.0	51.1	46.6
		MCDB	45.3	45.2	55.2	74.1	78.5	80.6	83.6	83.4	74.9	63.0	52.5	47.7
	2%	WB	40.6	40.9	45.6	56.3	64.1	67.5	70.6	69.0	63.0	56.4	48.5	43.3
		MCDB	41.9	42.7	50.6	66.6	73.5	77.1	80.0	78.0	69.7	59.5	49.9	44.4
	5%	WB	38.1	38.8	43.0	53.5	61.6	65.4	68.7	67.0	61.1	54.2	46.5	41.1
		MCDB	39.2	40.2	47.1	62.3	70.6	73.8	77.5	75.3	66.8	56.9	47.8	42.5
	10%	WB	36.2	36.9	40.6	50.9	59.2	63.2	66.9	65.1	59.2	52.2	44.8	38.8
		MCDB	37.1	38.4	43.7	58.2	66.6	70.9	74.6	72.9	64.2	54.7	46.1	39.9
Mean Daily Temperature Range	5% DB	MDBR	7.4	8.6	11.2	16.7	18.4	17.0	16.9	17.5	14.9	10.8	6.5	6.6
		MCDBR	6.9	8.2	17.2	24.2	24.0	22.6	22.7	23.9	20.4	14.6	8.5	6.9
		MCWBR	6.5	6.8	11.3	12.2	11.6	10.8	9.5	10.0	10.7	9.6	7.2	6.4
	5% WB	MCDBR	6.6	7.9	15.5	22.1	21.0	19.5	19.5	19.4	17.5	13.0	8.0	6.9
		MCWBR	6.3	7.0	11.0	11.8	11.3	10.4	9.4	9.5	10.1	9.4	7.1	6.6
Clear-Sky Solar Irradiance	taub	0.298	0.330	0.366	0.408	0.396	0.388	0.425	0.419	0.382	0.357	0.338	0.299	
	taud	2.198	2.181	2.212	2.196	2.293	2.343	2.261	2.273	2.349	2.398	2.282	2.235	
	Ebn at Noon	197	232	252	257	269	272	259	252	246	224	180	169	
	Edh at Noon	19	28	34	40	38	37	40	37	30	23	18	15	
All-Sky Solar Radiation	RadAvg	182	419	823	1293	1692	1761	1640	1374	954	469	171	115	
	RadStd	36	78	127	167	164	145	161	134	107	88	21	17	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	+0.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (18 neighbors)	+0.76	N/A	N/A	N/A	N/A	N/A	N/A	-266	N/A	N/A

Nomenclature: See separate page